3 Decisions Made for Form Digitisation Task

1. Extraction of text

Due to the PDF being text-based, I decided to proceed with the use of PyMuPDF that can read words, lines and drawing objects directly. The other option was the use of VLM or OCR pipelines, however that posed limitations due to the need to run models locally. The selected method was more efficient and faster to test. The trade-off is that unusual layouts may need new extraction rules later.

2. Human review before publishing

When the app runs the extraction process, the output is a draft and not final. I decided to allow for manual addition of new fields and editing of existing ones. This made the finalised draft to be more accurate to the uploaded form. This makes the system more reliable because PDF forms can vary a lot, and fully automatic field detection may not produce the most accurate results.

3. Normalised coordinates for PDF mapping

Field boxes are stored as coordinates relative to the page, set between 0 and 1 instead of raw screen pixels. This means the same template works regardless of browser preview size or zoom level. During export, the backend converts those normalized values back into PDF coordinates before writing the submitted values onto the original PDF.